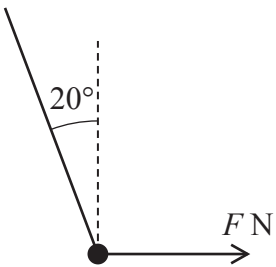
- The acceleration due to gravity is denoted by $g\,\text{m}\,\text{s}^{-2}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

3 A particle hangs at the end of a string. A horizontal force of magnitude F N acting on the particle holds it in equilibrium so that the string makes an angle of 20° with the vertical, as shown in the diagram. The tension in the string is 12 N.



(a) Find the value of F. **[2]**

(b) Find the mass of the particle. **[3]**